

Modeling and Simulation — Lesson 6

Synthesis: Choosing, Validating & Communicating Models

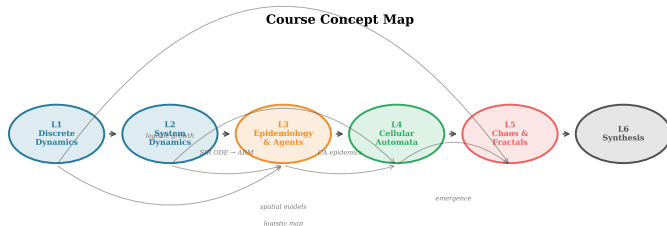
Václav B. Petrák

Faculty of Biomedical Engineering
Czech Technical University

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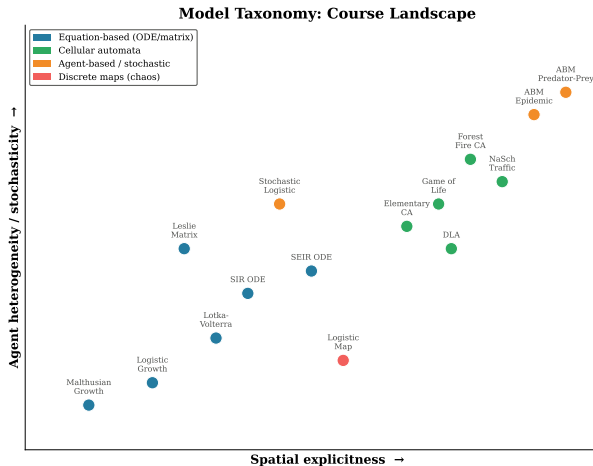
Model Taxonomy & Decision Framework

Looking Back: What Have We Built?



- ▶ Five lessons, each introducing a different **modelling paradigm**.
- ▶ Today we step back and ask: *when should I use which approach?*

The Modelling Landscape



- ▶ Two key axes: **spatial explicitness** and **agent heterogeneity**.
- ▶ Every model from the course occupies a different region.

Paradigm 1: Equation-Based Models

What: Ordinary differential equations (ODEs), difference equations, matrix models.

- ▶ Population is described by **aggregate variables** (S , I , R , N).
- ▶ Individuals are identical and well-mixed.
- ▶ Time can be continuous (ODE) or discrete (map / matrix).

Course examples:

- ▶ Malthusian & logistic growth (L1)
- ▶ Lotka-Volterra predator-prey (L2)
- ▶ SIR compartmental model (L3)
- ▶ Leslie matrix (L1)
- ▶ Logistic map (L5)

Paradigm 2: Cellular Automata

What: Grid of cells, each with a discrete state, updated by local rules.

- ▶ Space is **explicit**: a regular lattice.
- ▶ Time advances in discrete steps.
- ▶ **Emergence**: complex global patterns from simple local rules.
- ▶ No concept of individual “agents” — cells *are* the entities.

Course examples:

- ▶ Conway's Game of Life (L4)
- ▶ Forest fire model (L4)
- ▶ Elementary cellular automata (L4)
- ▶ Nagel-Schreckenberg traffic (project)

Paradigm 3: Agent-Based Models

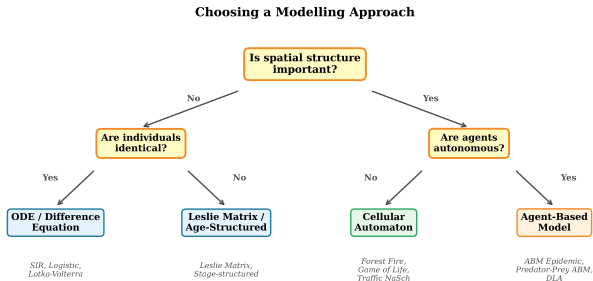
What: Individual agents with their own state, rules, and (often) position.

- ▶ Agents can be **heterogeneous**: different ages, speeds, infection status.
- ▶ Interactions depend on **proximity** or **contact networks**.
- ▶ Inherently **stochastic**: randomness in movement, infection, decisions.
- ▶ Most flexible — but hardest to analyse mathematically.

Course examples:

- ▶ Agent-based SIR epidemic (L3)
- ▶ Agent-based predator-prey (L2, project)
- ▶ Diffusion-limited aggregation (project)

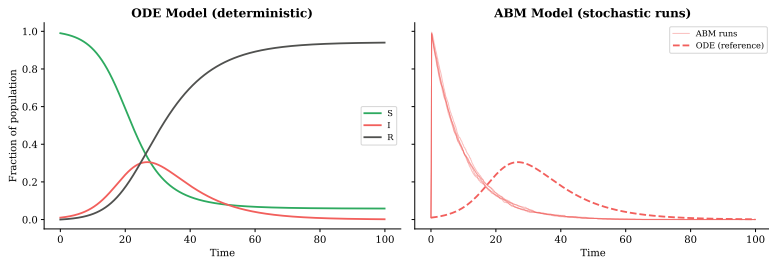
Choosing a Modelling Approach



† Add stochasticity to any approach: noise terms (SDE), random events (CA), random agent decisions (ABM)

Comparing ODE vs ABM: Same System, Different Lens

Equation-Based vs Agent-Based: SIR Epidemic



- ▶ ODE: smooth, deterministic, fast to compute.
- ▶ ABM: noisy, captures individual variation, computationally heavier.
- ▶ **Key insight:** the ODE is the “mean-field limit” of the ABM as $N \rightarrow \infty$.

Trade-offs Summary

	ODE / Matrix	Cellular Automaton	ABM
Spatial structure	No	Grid	Flexible
Heterogeneity	No	Limited	High
Stochasticity	Optional (SDE)	Rule-based	Inherent
Analytical results	Often	Rarely	Rarely
Computational cost	Low	Medium	High
Interpretability	High	Medium	Lower

- ▶ There is no single “best” approach — it depends on your **research question**.
- ▶ A good strategy: start simple (ODE), add complexity only when needed.

Validation, Sensitivity & Common Pitfalls

Why Validation Matters

- ▶ A model is a **simplification** of reality — it is always “wrong” in some sense.
- ▶ The question is: *is the model useful for the intended purpose?*
- ▶ George Box (1976): “**All models are wrong, but some are useful.**”

Three levels of checking:

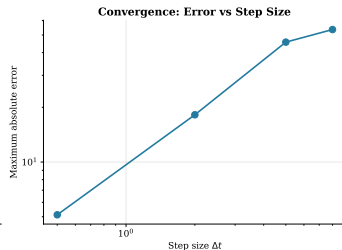
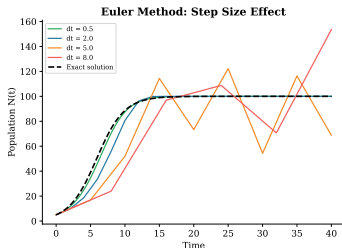
Verification Does the code correctly implement the equations?

Validation Does the model reproduce observed real-world behaviour?

Calibration Can we find parameter values that fit data well?

Verification: Is the Code Correct?

- **Conservation laws:** does $S + I + R = N$ hold at every step?
- **Limiting cases:** does the SIR reduce to exponential decay when $\beta = 0$?
- **Convergence:** does refining Δt improve the solution?
- **Comparison:** does the ABM match the ODE for large N ?

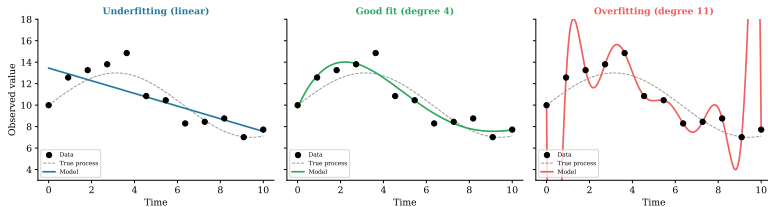


Validation: Does the Model Match Reality?

- ▶ Compare model output against **independent data** not used for fitting.
- ▶ Qualitative: does the model produce the right *shape* of behaviour?
 - S-curves, oscillations, extinction thresholds, etc.
- ▶ Quantitative: compute error metrics (R^2 , RMSE, AIC).
- ▶ **Important:** a model that fits training data perfectly may fail on new data.

The Overfitting Trap

Model Complexity: Finding the Right Balance

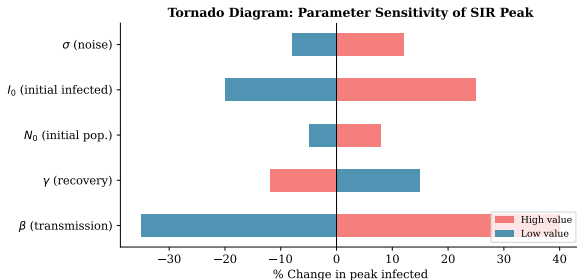


- ▶ **Underfitting:** model too simple, misses the pattern.
- ▶ **Overfitting:** model too complex, learns the noise.
- ▶ **Good fit:** captures the underlying process without chasing noise.
- ▶ **Rule of thumb:** *more parameters $\neq$ better model.*

Sensitivity Analysis

Question: How much does the output change when I vary a parameter?

- ▶ Vary one parameter at a time (OAT) while holding others fixed.
- ▶ Record a key output metric (e.g. peak infected, time to extinction).
- ▶ Visualise with a **tornado diagram**.



Sensitivity Analysis: Why It Matters

- ▶ Identifies **critical parameters** that drive the behaviour.
- ▶ Reveals which parameters need **careful measurement** vs. rough estimates.
- ▶ Helps communicate **uncertainty**: “If β increases by 10%, the peak doubles.”
- ▶ For your projects:
 - Pick 2–3 key parameters.
 - Sweep each through a plausible range.
 - Show how a key output responds.

Common Modelling Pitfalls

1. **Ignoring units.** Always track physical units (time⁻¹, individuals, etc.).
2. **Step size too large.** Euler's method can diverge or give wrong results.
3. **Too many free parameters.** If the model can fit anything, it explains nothing.
4. **No baseline comparison.** Always compare against a simpler model or null model.
5. **Cherry-picking runs.** In stochastic models, show *ensembles*, not single lucky trajectories.
6. **Confusing correlation with causation.** A good fit does not prove the mechanism is correct.

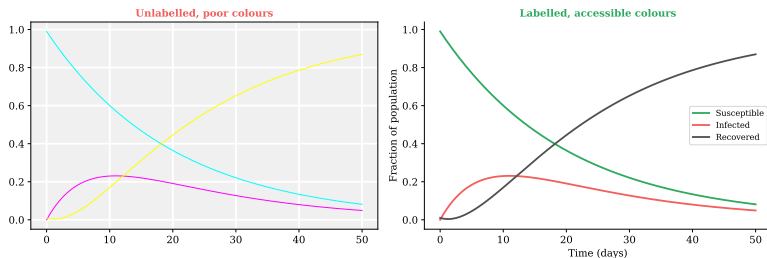
Stochastic Models: Ensemble Thinking

- ▶ A single ABM run tells you very little — it is **one sample** from many possible outcomes.
- ▶ Always run $N_{\text{runs}} \geq 20$ and report:
 - Mean trajectory
 - Confidence bands (e.g. 5th–95th percentile)
 - Distribution of key outcomes (peak size, final state)
- ▶ This is not a limitation — it is a **feature**: the spread tells you about risk and variability.

Communicating Results & Project Preparation

Visualisation Matters

Visualisation: Before & After



- ▶ Your plots are the **first thing** reviewers and audiences see.
- ▶ Bad visualisation can hide correct results; good visualisation can illuminate them.

Visualisation Checklist

Every figure in your report should have:

1. **Axis labels** with units (e.g. “Time (days)”, “Population N ”).
2. **Legend** or direct labels — the reader should never guess which line is which.
3. **Accessible colours** — avoid red/green only combinations (colour-blind readers).
4. **Caption** that describes what the reader should see.
5. **Consistent style** across all figures in the report.

Tools: `matplotlib` defaults are decent; use `plt.rcParams` to set a global style once.

Structuring Your Project Report

Recommended structure (4–6 pages + code):

1. **Introduction** What system are you modelling? Why is it interesting?
2. **Model Description** Equations, rules, parameters, assumptions. Include a diagram.
3. **Implementation** Key code design decisions. Not a code dump — highlight the important parts.
4. **Results** Figures with captions. What do the simulations show?
5. **Analysis** Sensitivity analysis, parameter sweeps, comparison to data or simpler models.
6. **Discussion** What did you learn? What are the limitations? What would you do next?

Writing About Models: Do's and Don'ts

Do:

- ▶ State all assumptions explicitly.
- ▶ Provide parameter values and their sources.
- ▶ Discuss what the model *cannot* capture.
- ▶ Compare against a baseline (simpler model, analytical solution, or data).

Don't:

- ▶ Claim the model “proves” something — models *suggest* or *are consistent with*.
- ▶ Present only the best-looking run out of many.
- ▶ Ignore parameters you did not vary.
- ▶ Skip units or dimensional analysis.

Presenting Simulations Live

For your final presentation (10–15 min):

1. **Hook** (1 min): What is the real-world problem?
2. **Model** (3 min): Show the key equations / rules. One diagram.
3. **Demo** (3 min): Run the simulation live or show an animation.
This is the highlight.
4. **Results** (3 min): 2–3 key figures. What did you discover?
5. **Reflection** (2 min): What worked? What surprised you?
What would you change?

Tip: prepare a **backup video/GIF** of the simulation in case the live demo fails.

Project Evaluation Criteria

Criterion	What we look for	Weight
Model correctness	Equations / rules are implemented correctly; code runs without errors	25%
Analysis depth	Parameter exploration, sensitivity analysis, comparison to baseline	25%
Visualisation	Clear, labelled, informative figures	20%
Report quality	Well-structured, assumptions stated, limitations discussed	15%
Presentation	Clear delivery, live demo, good slides	15%

Recap: The Modelling Workflow

1. **Define the question.** What behaviour do you want to understand or predict?
2. **Choose the paradigm.** ODE, CA, ABM — use the decision flowchart.
3. **Implement.** Start simple. Verify against known solutions.
4. **Explore.** Sweep parameters. Run ensembles. Compare to data.
5. **Validate.** Does it reproduce the qualitative / quantitative behaviour?
6. **Communicate.** Figures, report, presentation.

Good luck with your projects!